

Semantic Navigation in Latent Space for Early-Stage Industrial Design Exploration

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ABSTRACT

Generative models pack their variation into latent space—a compressed map of the designs they can produce. In most current workflows, designers engage with these spaces only indirectly, through isolated outputs or curated sets of images. Such indirect interaction limits their ability to perceive or intentionally explore the relationships between variations, especially during early ideation.

This thesis investigates how latent space might be reframed as a designer-facing medium for more open and deliberate exploration. It begins by surveying current methods of representing latent spaces—such as semantic axes, spatial mappings, and tagging interfaces—to establish a structured understanding of how interaction with these spaces is currently approached. Building on this, I will prototype several interface strategies designed to present latent structure in ways that are both meaningful and accessible to designers.

Rather than proposing a comprehensive solution, this research focuses on identifying specific, practical interaction techniques that support designers' iterative and exploratory workflows. By bridging machine learning representations with parametric design logics, the aim is to uncover ways for designers to navigate generative variation more intuitively and intentionally.

INTRODUCTION

Prompt-Based Design and the Problem of Latent Inaccessibility

In today's generative workflows, tools like Stable Diffusion or Midjourney have opened up striking new possibilities for visual ideation. But for designers, especially in early-stage industrial design, these systems often resemble batch generators—you type a prompt, receive a grid of outputs, and try to select or remix what's interesting. While impressive, this process remains largely detached from a designer's ability to understand or steer variation intentionally. Outputs are generated as isolated samples; designers must infer how prompt phrasing relates to form changes, with little feedback from the model itself.

This indirect interaction introduces a disconnect between intention and output. Because prompt-based systems offer limited feedback on how specific inputs relate to underlying variations, designers often need to experiment through trial and error. During early-stage ideation, this can make it more difficult to iteratively adjust or steer form development in a controlled or interpretable way. While latent space encodes a wide range of potential design outcomes, its structure remains difficult to access or understand directly—especially for designers aiming to explore variation semantically or spatially. Designers often rely on trial-and-error prompting to generate sets of outputs, with limited feedback about how specific input phrasing relates to variation (see Figure 1).

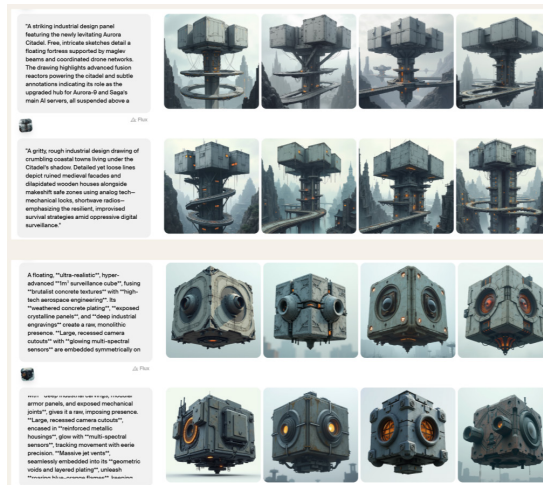


Figure 1. Prompt-based exploration using Krea AI. Designers must infer relationships between prompt text and form variation from generated image sets.

Structure-Driven Exploration and the Uncharted Map of Latent Space

Meanwhile, a few tools already let designers poke directly at that structure. Visualizations of latent interpolation paths, 2D scatterplots of outputs, and dimensionality reduction techniques like UMAP or t-SNE suggest that latent space encodes rich variation in shape, proportion, and semantics. For example, Maesumi et al.'s explorable mesh deformation subspaces demonstrate how latent variation can be made legible through visual mappings (Figure 2) [1]. Similarly, ChairGAN's scatterplot interface spatializes design outputs to highlight clustered relationships—yet these remain detached from intuitive control or semantic labeling (Figure 3) [2].

These efforts affirm that latent variation has structure—but also reveal persistent barriers to making that structure usable. Most visualizations are only indirectly understandable: axes are rarely defined in designer-relevant terms, and small movements in latent space can yield unpredictable changes. Semantic models like CLIP offer potential pathways by embedding text-image alignments (e.g., “minimal” to “expressive”), but such mappings are rarely navigable or adjustable within actual design workflows. This research begins from that gap: rather than building new generative models, it asks how interface-level interventions might help designers explore latent variation with more clarity, flexibility, and control.

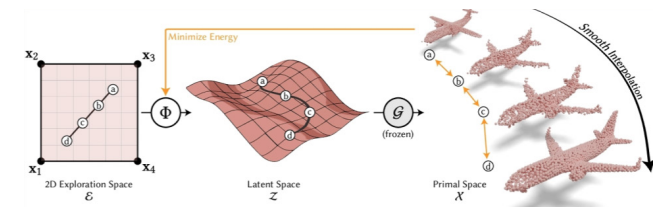


Figure 2. Explorable mesh deformation subspaces from Maesumi et al. (2023), showing transitions between forms along structured latent paths.

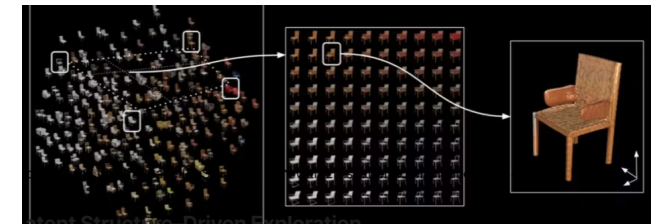


Figure 3. Latent space scatterplot navigation from ChairGAN BLS interface, showing semantically clustered design outputs.

TERRITORY MAP: SITUATING THE RESEARCH FOCUS

Between Systems: My Background and Research Interest

My own journey begins in rule-based design systems like Rhino and Grasshopper, where explicit parametric logic governs spatial behavior. These systems taught me how to structure variation, manage constraints, and reason through spatial dependencies. Over time, this shaped how I learned to design—not just as a process of form-making, but as a way of constructing systems.

Coming to CMU, I set out to encounter a different paradigm. Generative methods such as GANs and diffusion models offer compelling expressive potential, but many operate through abstract representations—dense vector spaces, training priors, or multimodal embeddings—that aren't immediately intuitive within early-stage design workflows. In practice, it can be difficult to grasp what types of variation are encoded, how differences relate to one another, or what specific directions in latent space correspond to semantically.

Rather than view these challenges as shortcomings, I see them as opportunities. My interest is not in replacing rule-based systems or simply sampling from generative ones, but in investigating the middle ground: interfaces that help designers seek, label, steer, or reshape variation in productive ways. This includes scatterplot navigation, semantic sliders, keyframe interpolation, and clustering techniques—methods that begin to turn latent space into a navigable design surface.

This aligns with Mario Carpo's (2024) framing of

“rice or pasta?”—a metaphor for the divide between deterministic expert systems and probabilistic imitation-based AI. [3] What I'm drawn to is neither pole, but the in-between space: design workflows that engage with variation while remaining interpretable and responsive to designer intent. With a background in rule-based parametric tools and a growing interest in data visualization, I see latent interfaces as a promising pathway—though not the only one—for making the inner workings of generative models more accessible. My attention is on how latent vectors, often buried deep within model architectures, might be re-surfaced through visualization and interaction patterns that resonate with design logic. This is not to suggest that visualizing latent space is a universal solution—but it offers one way to reframe these abstract representations into something iterative, and human-centered, bridging the expressive affordances of generative AI with the structured thinking of design workflows.

Mapping the Terrain: A Fragmented Ecosystem and Where I Focus

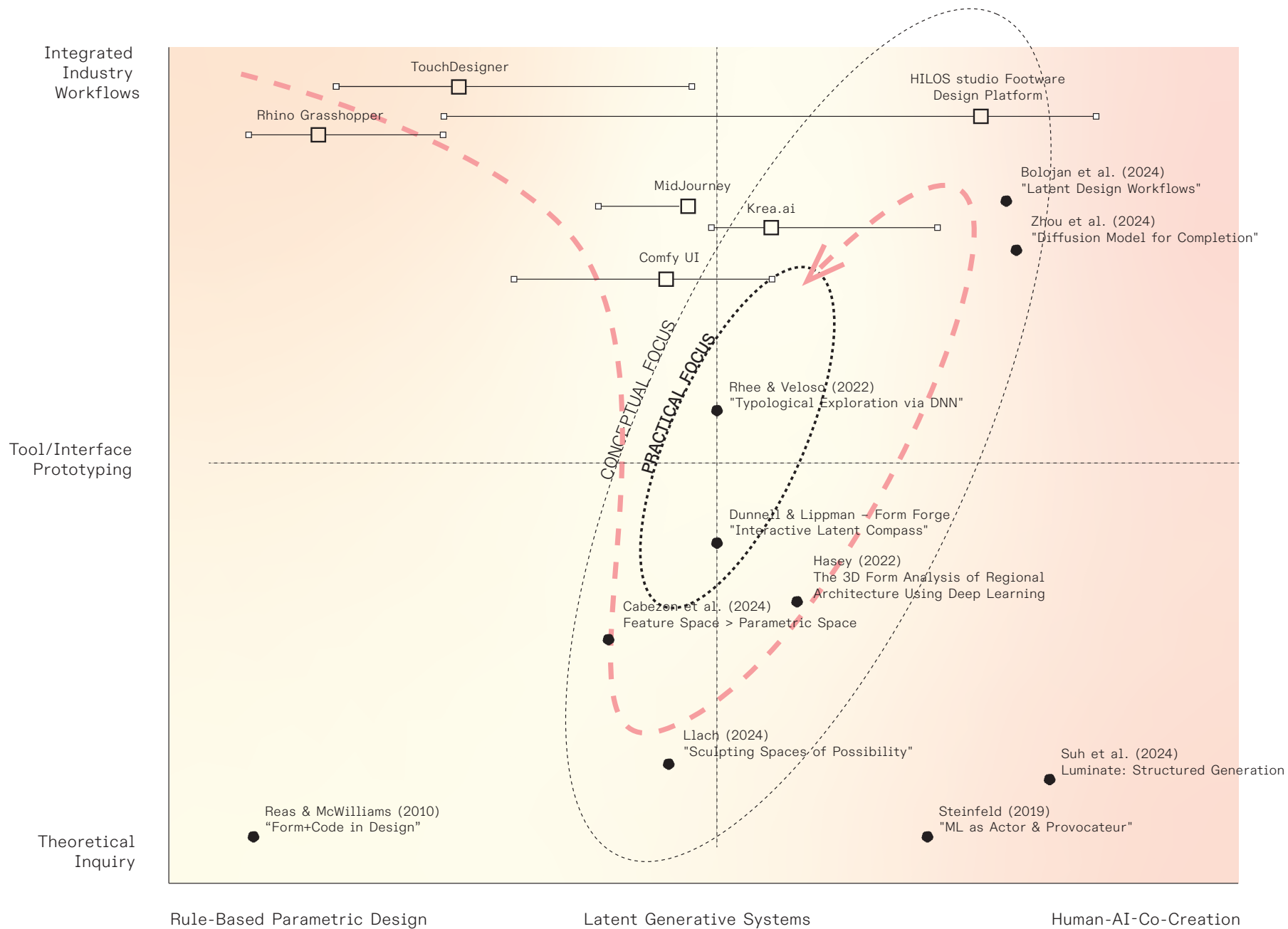
The territory map on the next page attempts to give this landscape a form. It charts two intersecting axes: from rule-based parametric systems to human-AI co-creation, and from theoretical inquiry to workflow-integrated tools. Together, they depict a fragmented ecosystem in which traditional modeling platforms, experimental generative systems, and speculative interface research often coexist—but rarely align in ways that meaningfully support designers' exploratory workflows.

The shaded region marks a lightly populated but highly relevant area of interest: latent interface tools

that are both semantically searchable and attuned to design logic. For example, DeepRise [4] enables latent manipulation for architectural massing, offering sliders and interpolation modes that support user-driven variation—but these are limited to narrow typologies and lack broader semantic steerability. Form Forge [5] employs a radial compass to reveal specific latent variables, yet its axes remain fixed, without customization or semantic alignment from the designer's perspective.

Meanwhile, related work in explainable AI—such as Luminate by Suh et al. [6], which structures outputs using language models, or Zhou et al. [7], who propose graph-guided completions—aims to make generative systems more legible. However, these tools primarily serve technical understanding and model auditability, not designer-centered ideation.

This thesis builds on these efforts but addresses a distinct gap: how to surface and interact with latent structure in ways that align with early-stage design thinking. It investigates how interface-level structures—semantic sliders, tag-based filters, and exploratory maps—might help designers navigate, reinterpret, and direct variation within generative systems. The ambition is not to decode latent space fully, but to make it meaningfully navigable: a space of possibility shaped as much by designer intent as by model architecture.



□ Established Design Tools

● Research Paper of Interest

— Path of Approaching Research with Personal Experience



Estimated Research Saturation by Field**

"Gradient background reflects my interpreted density of research activity across the space, highlighting gaps in latent design workflows and mid-spectrum integrations."

STATE OF THE ART & GAPS

Existing Interfaces for Exploring Latent Variation

Several research tools have experimented with interfaces for navigating latent space—using scatterplots, tagging, radial layouts, or interpolation paths to surface structure behind generative outputs. These tools offer valuable insights and establish a baseline of interaction patterns that this thesis builds upon. Rather than treating them as incomplete, I view them as critical precedents for developing a taxonomy of designer-facing strategies. However, most still fall short in enabling real-time semantic steering, integrating with sketch-like workflows, or revealing how designers actually traverse variation. This research proposes an interface that makes latent structure both visible and steerable—shifting variation from something passively interpreted to something actively explored.

Form Forge (Dunnell & Lippman, 2024) introduces a radial interface where a compass of tick marks surrounds a StyleGAN-generated form. Each tick maps to an explicit latent variable, enabling users to incrementally adjust individual dimensions and observe corresponding changes in real time. The approach offers a tangible sense of steering form generation, but focuses on local edits without providing a global view of the design space (Figure 4) [5].

Browsing the Latent Space (BLS) (Cabezon et al., 2024), built on ChairGAN, spatializes generated chairs into a 2D scatterplot based on latent similarity. Users can move between nearby outputs and interpolate between samples, revealing clustered design families. However, axes are fixed by the projection algorithm; users cannot define their own semantic directions or

reorient the map (Figure 5) [2].

Runway's latent path editor supports media generation by letting users define keyframes and interpolate across latent transitions. Graph nodes represent visual outputs (e.g., video frames), and edges define how outputs morph between points. While powerful for motion and storytelling, it lacks tools for fine-tuning latent axes or surfacing semantic relationships (Figure 6) [8].

DeepRise (Rhee & Veloso, 2022) integrates latent-space interaction into architectural massing studies. Its interface includes three exploration modes—random, vector-based synthesis, and interpolation—allowing designers to scroll, adjust sliders, or blend forms. Yet the tool remains domain-specific and focuses on typological variation, without general tools for declaring or labeling semantic axes (Figure 7) [4].

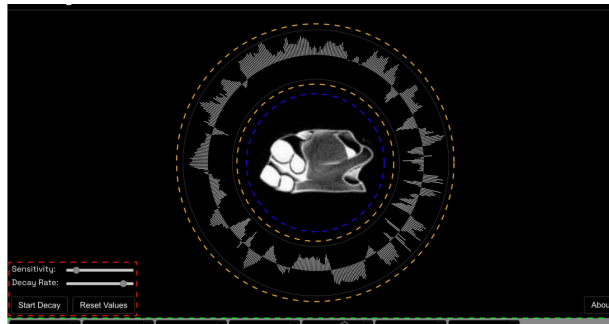


Figure 4. Form Forge interface showing radial tick-based control of StyleGAN variables for architectural massing (Dunnell & Lippman, 2024).

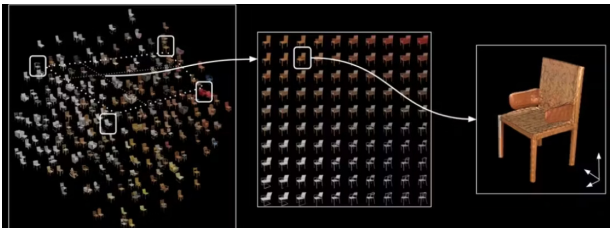


Figure 5. BLS interface displaying ChairGAN outputs in a scatterplot. Users navigate via proximity, but cannot redefine axes (Cabezon et al., 2024).



Figure 6. Runway path tool for latent interpolation in video generation, using graph-based transitions between keyframes (Runway, 2024).

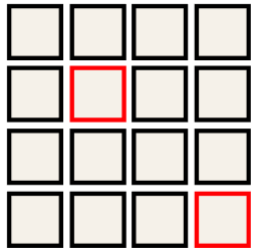


Figure 7. DeepRise interface with three latent control modes for high-rise form generation: exploration, synthesis, and interpolation (Rhee & Veloso, 2022).

BUILDING A TAXONOMY OF INTERACTION MODES

Following the analysis of precedent tools, this section organizes a broader set of latent space interaction strategies into five common categories. While prior tools such as Form Forge, DeepRise, or BLS explored key patterns—radial control, interpolation, or spatial clustering—many other tools across platforms like MidJourney, Runway, or GANSpace contribute additional insights that help surface a more complete picture. This emerging taxonomy offers a framework for describing how designers currently encounter, navigate, and interpret generative variation. Each mode below represents a recurring interaction strategy, categorized not by technical implementation, but by how users conceptually relate to variation and semantic meaning in latent space.

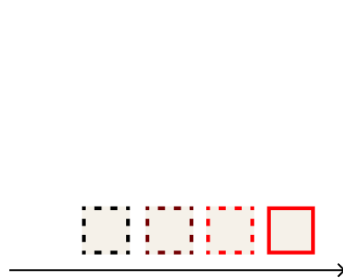
These categories are not exhaustive, but form a working schema that this thesis builds upon. By identifying and critically evaluating these interaction modes, I aim to surface latent patterns in current design interfaces and explore how they may be adapted or extended to support early-stage, designer-driven workflows. Going forward, this taxonomy will inform both analytical studies and prototyping efforts, helping to assess how each mode supports semantic control, spatial reasoning and iterative design within latent space.



Raw Batch

Examples: MidJourney, DALL·E 2

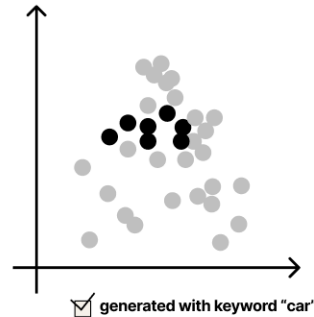
Outputs are presented as unordered grids, with no explicit structure. Designers rely on intuition to interpret relationships, limiting control or iteration beyond prompt tweaking.



1D Axis Interpolation

Examples: Runway ML, StyleGAN interpolation demos

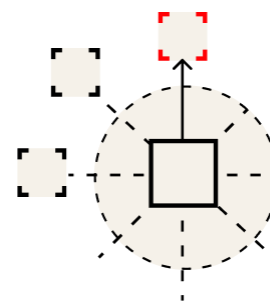
Users blend between two points in latent space to observe transitions. This reveals some underlying continuity but offers limited dimensional control or semantic orientation.



Semantic Filters

Examples: Lexica, CLIP Interrogator, Krea.ai

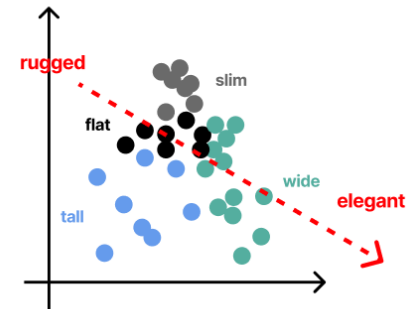
Keyword prompts or tags narrow results, often after generation. While useful for categorization, they don't show how outputs relate or evolve structurally in latent space.



Radial Layouts

Example: Form Forge (MIT Media Lab)

Radial layouts map latent variables to adjustable ticks around the object. This enables direct manipulation of certain dimensions but lacks holistic spatial context.



Axes + Cluster Overlay

Examples: GANSpace, SemanticGAN, StyleGAN Explorer

These interfaces visualize outputs on semantic axes with clustered regions, helping users identify patterns and navigate variation more intentionally.

POSITIONING THE PROTOTYPE: WORKFLOW AND DESIGN SCOPE

Designing for Workflow Insertion in Industrial Design

This prototype begins within early-stage industrial design, where standalone objects and semantically legible traits (e.g., minimal vs. expressive) offer a clear setting for exploring structured variation. These workflows favor visual, iterative reasoning—making them ideal for testing latent space navigation tools. The choice is also strategic: an upcoming footwear design internship offers access to applied generative tools in production settings. This creates an opportunity to observe how such tools might integrate with actual workflows—not as general solutions, but as exploratory layers within established design pipelines.

Scoping the Prototype: From Concept to Controlled Implementation

This research does not aim to create novel generative models or optimize output realism. Instead, it focuses on how semantic structures and interaction patterns can help designers explore latent variation more intentionally. Rather than batch-generating outputs for passive review, the prototype encourages users to define or adjust axes of variation (e.g., “elegant → rugged”) and visually navigate the generative space.

To keep development and evaluation focused, the initial scope is constrained to 2D image generation, using well-established models like StyleGAN or Stable Diffusion in combination with semantic embedding via CLIP. This allows for lightweight experimentation while providing meaningful visual outcomes.

The prototype will use 1–2 semantic axes and test a simple UI structure—such as sliders, preview panels, or scatterplot maps—to support interpretability.

The interface will be embedded in the early exploration phase of the design process, where generative tools can support loose ideation rather than refinement. (Figure 8) Whether this prototype supports iterative feedback or more one-shot exploration will depend on further understanding of designer behavior in relevant workflows—both through testing and situated research. Ultimately, this project seeks to examine how early-semantic structuring can be inserted not just as a model output viewer, but as a thinking interface—one that aligns generative variation with the fluid reasoning and evolving intuition that characterizes creative design work.

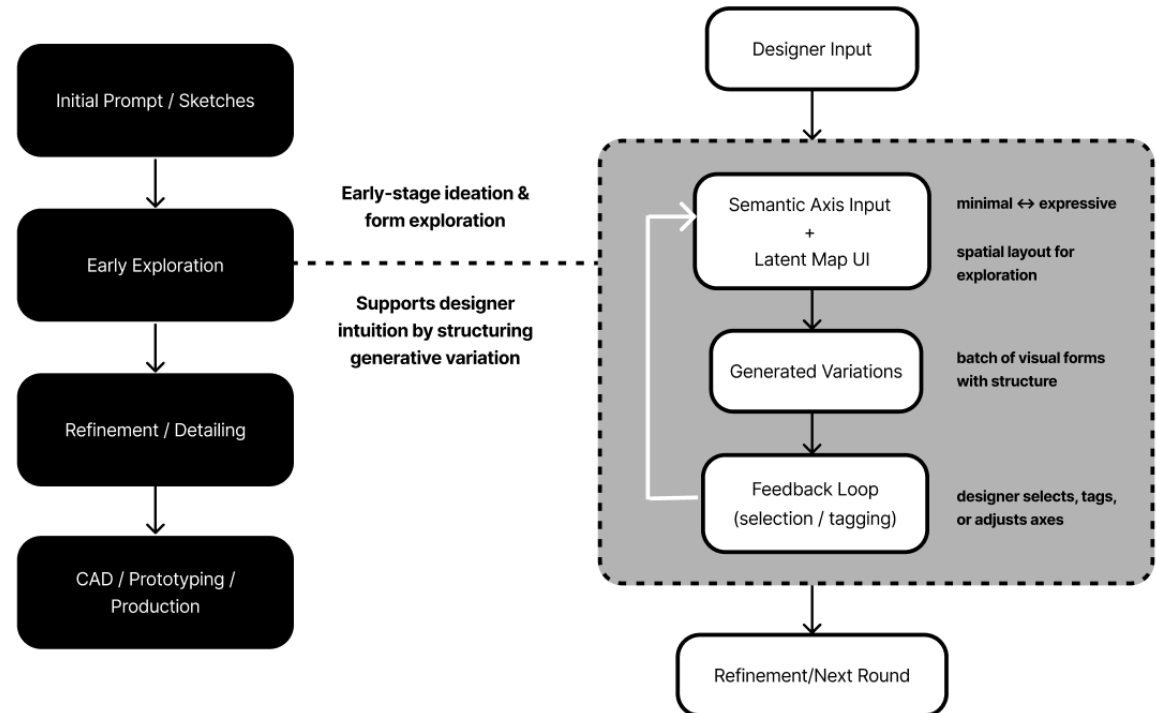


Figure 8. Workflow diagram showing how the prototype inserts into early-stage industrial design. Designers input a semantic axis (e.g., elegant ↔ rugged), then explore structured latent variation through a simple UI, selecting or refining outputs in a feedback loop. The prototype supports ideation by making variation more navigable and interpretable.

TESTING THE INTERFACE: METHODS AND EVALUATION

Purpose and Framing

Rather than optimizing for realism or benchmarking against the outputs of advanced models, this thesis investigates how designers interact with latent variation—specifically through the lens of semantic structure. The focus is on prototyping early-interface interventions that reveal, rather than obscure, the nature of variation: sliders aligned with conceptual axes, tagging and feedback mechanisms, or spatial layouts that make structure navigable.

The prototype will leverage pre-trained diffusion or GAN-based models (e.g., Stable Diffusion + CLIP), but the specific model choice will be determined following further technical review—including potential transformer-based alternatives. The ambition isn’t to produce new forms of content, but to probe how interface arrangement affects the interpretability of variation, supports semantic iteration, and enables designers to navigate these spaces with greater deliberateness. Evaluation, accordingly, won’t hinge on output fidelity, but instead on how clearly variation is surfaced, how intentionally it can be explored, and how much agency users perceive in the process.

Drawing on Kyle Steinfeld’s argument that speculative artefacts can act as “significant others” that provoke dialogue with computational systems rather than verify them [9], this thesis treats its interface probes as designed provocateurs—not finished products. By foregrounding the frictions and ambiguities that arise when designers steer latent variation, each prototype is meant to surface hidden assumptions about agency, authorship, and control, inviting critical reflection on how machine-learning models participate in early-stage ideation. The ambition, therefore, is less to ship a polished tool than to generate situated insight into the mutual shaping of designer intention and model behaviour.

Evaluation Strategy and Development Trajectory

To assess the effectiveness of interaction modes, three core dimensions will be tracked (see Figure 9): semantic clarity (how legible and differentiated the variation appears to users), exploration patterns (how users traverse and interact with the latent space), and perceived control (the degree to which users feel they can direct outcomes). These will be tested through a combination of A/B interface mockups, movement tracking, and slider or tagging tasks. Where appropriate, Wizard-of-Oz setups may be employed in lieu of full backend implementation, allowing insight into interaction patterns without overengineering the tool.

The prototype will support side-by-side comparison of multiple UI configurations—including scatter-plots, sliders, and clustered or tag-driven layouts—each probing different interaction behaviors. Early tests will likely rely on synthetic or pre-processed datasets rather than industry-specific data such as footwear. However, insights gathered during my upcoming internship will offer a grounded reference point for later refinement, helping align interface experiments with real-world creative pipelines.

TEST FOCUS	METHODS	MEASUREMENT
SEMANTIC CLARITY	A/B Interface Mockups	Clarity of interpretation
EXPLORATION PATTERNS	Designer Movement Tracking	Breadth of exploration
PERCEIVED CONTROL	Tagging + sliders	Sense of agency

Figure 9. Table summarizing test focus, interface methods, and evaluation criteria.

TIMELINE & NEXT STEPS

Summer 2025

— Foundational Research & Algorithmic Exploration

This phase will focus on mapping the landscape of existing interface metaphors—including sliders, semantic filters, latent scatterplots, and clustering overlays—to examine how current tools have made latent variation interpretable. These references will serve not only as inspiration but as a basis for organizing a taxonomy of interaction strategies, which will shape the design language and user logic of my own prototype.

In parallel, I will conduct technical trials to experiment with latent structure mapping using pre-trained generative models such as Stable Diffusion + CLIP. I plan to generate embeddings and visualize them through dimensionality reduction techniques like UMAP or t-SNE, observing how forms, meanings, or attributes cluster in space. These mappings will help surface emergent semantic axes and prompt hypotheses around which types of structure are most meaningful to designers.

During this period of summer internship, I will conduct embedded observation at a generative AI software company supporting footwear design workflows. By examining how designers interact with AI-driven tools in practice, I'll identify gaps between system affordances and design intent—insights that will help guide how semantic latent navigation might better align with real-world workflows.

Fall 2025

— Initial Testing & Prototyping

In the fall, the project will shift toward prototyping and evaluation. I will select one or more pre-trained models—likely StyleGAN, Stable Diffusion, or Transformer-based models (pending technical feasibility and dataset compatibility)—to serve as the generative engine. Model choice will depend on their openness to semantic conditioning and the clarity of latent representation.

Low-fidelity, functional UI mockups will be developed, incorporating features such as semantic axis sliders, preview panels, tagging mechanisms, or scatterplot-based navigation. The core goal will be to test different interaction modalities and observe how they support or hinder designer engagement with generative variation. These tests will center around three dimensions of interest: semantic clarity, exploration patterns, and perceived control.

Evaluation will be conducted through informal user probes, including A/B comparisons, tagging tasks, and logging of navigation behaviors. Where necessary, I will use Wizard-of-Oz techniques to simulate system responses—allowing the focus to remain on interaction flow and perception, rather than model speed or output fidelity. These evaluations will help assess whether interface adjustments can meaningfully shift how designers explore, sort, or interpret generated outputs.

Spring 2026

— Refinement & Framing

Spring will focus on iterating and framing the prototype, refining its logic and interface based on user feedback from the fall. Rather than improving the underlying model, the emphasis will remain on the surface-level structuring of variation: how designers can steer, label, or resurface content in meaningful ways, especially during early-stage ideation.

This will also be the phase of positioning the project in a broader discourse—articulating how semantic navigation in latent spaces sits within generative design, not as a technical breakthrough, but as a designer-facing interpretive method. I aim to show that these tools are not just output viewers, but interfaces for thought, capable of revealing patterns and surfacing intention.

Finally, I will begin thesis documentation and communication planning, weaving together situated insights, technical discoveries, and critical reflections into a cohesive narrative. The resulting work will communicate both the speculative value and practical affordances of latent space navigation—shaped by interaction, shaped by design.

WHY ME: POSITIONING, SKILLS, AND PREPARATION

A Situated Perspective

This research emerges from a hybrid background in design and computational systems. My training began with rule-based platforms like Grasshopper, where I learned to structure spatial logic, manage constraints, and reason through parametric relationships. This approach shaped how I think about design—not only as form-making, but as constructing and navigating structured variation.

At CMU, I have expanded my focus to include machine learning methods such as diffusion and transformer models. These techniques offer expressive possibilities but often encode variation in ways that are less immediately interpretable. This project examines how those generative outputs might be better aligned with the forms of reasoning and decision-making common in design workflows.

Rather than proposing new models or complete tools, this thesis focuses on how designers might work with variation more intentionally—through interfaces that clarify structure, support comparison, or aid in semantic navigation. This direction is informed by Daniel Cardoso Llach's concept of sculpting spaces of possibility, which frames AI not as a deterministic generator, but as a design tool that can reflect and extend creative agency [10]. In doing so, the research seeks to improve how designers interact with existing generative systems during early-stage ideation—where tools should support exploratory thinking, rather than final output selection.

Skills and Preparation

This thesis draws on a mixed foundation of design tools, programming, and growing ML fluency:

- In computational design, I have extensive experience with Grasshopper and parametric workflows, giving me a strong base for thinking through structured variation.
- In machine learning, I've completed CMU's Intro to Deep Learning course and can conduct inference, build simple models, and understand architectural principles—though I aim to further develop skills in fine-tuning, model selection, and deployment.
- On the software side, I've worked with web-based front-end frameworks (Svelte, some React) and data/graphics libraries (Three.js, D3). To bridge interaction and visualization, I will continue improving proficiency in JavaScript and UI design.
- I am also pursuing exposure to HCI methods via coursework and collaborative projects, aiming to complement my intuitive design sensibility with structured user testing frameworks.

Embedded Industry Context

My upcoming internship at HILOS, a startup developing generative AI pipelines for footwear design, offers a grounded site for observation. Rather than extracting case studies post-hoc, I will engage in embedded inquiry, documenting how AI systems are currently integrated into design workflows, and how designers interact with them—both effectively and critically.

This real-world experience will help calibrate the assumptions built into my prototype. It will also inform the design of interface affordances, based not just on technical capability, but on actual decision-making patterns, exploratory behavior, and moments of friction observed in industry.

Ultimately, this project positions me not as a domain expert in ML or a traditional UX researcher, but as someone situated across disciplines—interested in prototyping, interpreting, and interrogating the role of latent variation in the future of design tooling.

DISCUSSION, REFLECTION, AND CONCLUSIONS

This thesis addresses a focused yet timely challenge: how to reframe latent space as a design-attuned medium rather than a residual byproduct of machine learning systems. This specific focus offers conceptual clarity but also brings certain constraints—chiefly, the need to more rigorously define how latent space differs from broader generative AI discourse, and why its structural characteristics are uniquely relevant to early-stage design ideation.

A key area for further development lies in deepening my foundational understanding of how latent spaces are constructed—across GANs, diffusion models, and transformer-based architectures—and how those constructions have evolved within the field. Beyond their utilitarian role in generating images or interpolations, I aim to investigate how latent spaces have been theoretically framed in machine learning literature, and how those framings intersect—or misalign—with the epistemologies and workflows of design. This will help better articulate and defend why I've chosen this particular space as the core of inquiry.

Given the speed of change in generative AI, it is also vital to remain attuned to new developments—emerging tools, platforms, and interaction paradigms. My upcoming internship at a GenAI-focused software company working on footwear design pipelines offers a valuable opportunity to observe firsthand how generative tools are embedded into real-world workflows. This kind of situated research enables me to focus not just on what is technically possible, but on what is meaningfully usable by designers—what fits their mental models, pace of iteration, and forms of reasoning.

Previous feedback has also pointed to an important practical consideration: even without training new models, any interface or interaction prototype still depends on accessible, representative data. I'll remain flexible in this area, identifying public datasets or using lightweight simulation to build and test UI concepts. Wizard-of-Oz techniques may be used where full integration isn't necessary, especially in early testing phases where designer interaction—rather than model performance—is the focus. This leads to another important question: what kind of testing actually leads to insight? My emphasis is not on benchmarking or realism, but on evaluating sensemaking. Does the interface clarify variation? Does it support active traversal, reinterpretation, and feedback? Success will be measured less by what the model can produce, and more by how designers interpret and orient themselves within that generative structure.

In essence, this thesis operates in a unique liminal space—between parametric logic and generative systems, between technical abstraction and designer intuition. **Figure 10** (right) encapsulates this positioning. It visualizes the convergence of personal background and external observations, mapping the core question that drives this inquiry: how can latent space be made more legible, navigable, and open to designer control?

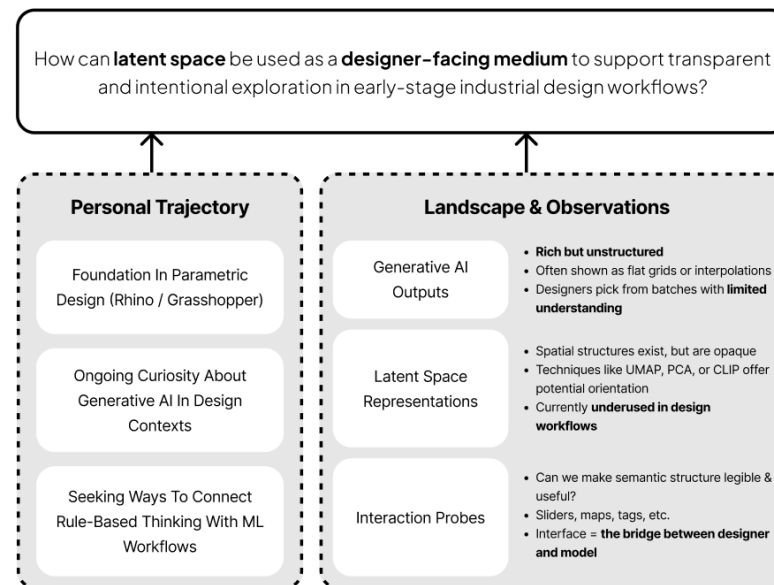


Figure 10. Summary diagram showing the convergence of personal trajectory and external landscape observations, framing the central research inquiry: how latent space can serve as a designer-facing medium for intentional, transparent exploration. This diagram distills the motivations, gaps, and interface opportunities that shape the scope of the thesis.

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